National Academic Test Program
โครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.)

การแข่งขันสอบวัดทักษะวิชาการระดับชาติ ประจำปี พ.ศ. 2565

(รอบสุดยอดอัจฉริยภาพระดับประเทศ)

วิชาภาษาอังกฤษ ระดับชั้นมัธยมศึกษาตอนปลาย

รหัสวิชา อ405 ภาษาอังกฤษ

รหัสชุดข้อสอบ 405007

ชื่อ.....นามสกุล.....รหัสวิชา.....

ระดับชั้น.....ห้องสอบ.....วันที่สอบ.....สถานที่สอบ.....

คำชี้แจงการสอบ

รายละเอียดข้อสอบ	**ข้อสอบฉบับนี้มี 10 หน้า จำนวน 50 ข้อ ระยะเวลา 90 นาที**		
วิธีการตอบ	**ให้ใช้ดินสอ 2B ระบายในวงกลมที่เป็นคำตอบในกระดาษคำตอบ หรือ นำโทรศัพท์มือถือมาสแกน QR Code เพื่อกรอกและยืนยันคำตอบในระบบ**		
เกณฑ์การให้คะแนน	**ส่วน Expression**	**ข้อ 1-10**	**ข้อละ 1.6 คะแนน**
(คะแนนเต็ม 100 คะแนน)	**ส่วน Reading Comprehension**	**ข้อ 11-25**	**ข้อละ 2.4 คะแนน**
	ส่วน Structure	**ข้อ 26-40**	**ข้อละ 2.2 คะแนน**
	ส่วน Vocabulary	**ข้อ 41-50**	**ข้อละ 1.5 คะแนน**

ข้อปฏิบัติในการสอบ

1. ให้ผู้เข้าสอบปฏิบัติตามระเบียบสำนักงานโครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.) ภาคีเครือข่ายโรงเรียนทั่วประเทศ และสถาบันอีซี่ลิชประเทศไทย ว่าด้วยแนวทางปฏิบัติเกี่ยวกับการดำเนินการทดสอบ พ.ศ. 2560 อย่างเคร่งครัด
2. สำหรับสนามสอบปิด ให้ผู้เข้าสอบระบายคำตอบลงในกระดาษคำตอบด้วยดินสอ 2B อย่างถูกต้องและชัดเจน
3. สำหรับสนามสอบออนไลน์ เมื่อผู้เข้าสอบนำโทรศัพท์มือถือหรืออุปกรณ์อิเล็กทรอนิกส์มาแสกน QR Code เพื่อเข้าสู่แบบฟอร์มกระดาษคำตอบ กรอกหรือเลือกคำตอบให้ถูกต้องและครบถ้วน โดยเปิดกล้อง ZOOM ยืนยันตัวตนไว้จนกว่าจะยืนยันกระดาษคำตอบเสร็จ และบันทึกภาพหน้าจอเพื่อเป็นหลักฐานหลังกดส่งยืนยันคำตอบและปิดกล้อง

หากผู้เข้าสอบฝ่าฝืนข้อปฏิบัติโครงการ สวช. หรือมีการทุจริต จะมีการดำเนินการ ดังนี้

1. ไม่ประกาศผลการสอบจากการทดสอบดังกล่าว
2. แจ้งไปยังโรงเรียนต้นสังกัดของนักเรียน ผู้ดูแลของผู้สอบเพื่อดำเนินการตักเตือนและกล่าวโทษ
3. แจ้งไปยังผู้บริหารสถานศึกษาที่เป็นภาคีเครือข่าย สวช. เพื่อพิจารณารับเข้าศึกษา

เอกสารนี้เป็นลิขสิทธิ์ของสำนักงานโครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.)
มูลนิธิโครงการสอบวัดทักษะวิชาการและโอลิมปิกนานาชาติ
การทำซ้ำหรือดัดแปลงหรือเผยแพร่เอกสารฉบับนี้จะถูกดำเนินคดีตามกฎหมาย

Part I: Expression (Items 1 – 10)
Choose the best answer.

Situation: Selecting a school representative

Principal: I've called you here to discuss about the selection of a new representative for the national competition. _____(1)_____, so far there are only two candidates, Benjamin and Cartoon. First, I'd like to hear from your perspectives, both_____(2)_____, about Benjamin's qualifications for this responsibility.

Matys: Actually, I have a thought that Benjamin is the perfect person for this position. He's been with the school for three years, so he knows the school _____(3)_____. He's quite bossy and serious; plus, he gets along well with everybody.

Giacomo: _____(4)_____ In my point of view, if he becomes a representative, nothing will be outstanding. I mean, we wouldn't see anything creative – just the opposite – we'd keep on implementing the same presentation as usual.

Elena: He isn't known for thinking outside the frame. Furthermore, we'd start to see our popular vote and student base fall off, simply because our competitors have active new people and creative presentation. _____(5)_____

Puke: I have to agree with Giacomo and Elena on this one. Primarily, Benjamin is too conservative; yet, he'd barely stimulate any creativity among the team members.

Principal: Anyone else want to _____(6)_____? OK. Then let's discuss about Cartoon. What do you guys think?

Matys: Listen, if Cartoon gets this position, it will be a catastrophe for the school. In the first place, her sophisticated degree isn't worth squat because she has zero job experience. In the meantime, we don't even know her very well. _____(7)_____, she's only been here since December, and on top of that, we all know that she doesn't have that effective presentation skill.

Puke: Really, I've heard that her colleagues think she's _____(8)_____, like she thinks she's the queen of the school. She's not exactly popular with the other team members. I don't think they'd be happy with her as the representative.

Eleni: Well, _____(9)_____, I have to say that I consider her as a very wise and competent person. Still, I recognize that she lacks experience. And now that you tell me that her personality could cause friction among the team members, then I'll go along with your decision in this case.

Principal: Obviously, we haven't found the ideal person to handle this position. We may have to look outside the team, which I don't practically want to do. We'll meet here tomorrow at the same time. I'll expect your suggestions – and _____(10)_____

1.
1. As it is shown
2. As you acknowledge
3. As long as it is stated
4. Particularly, as it is
5. According to the statement that

2.
1. pro and con
2. back and forth
3. black and white
4. work life balance
5. his and her

3. 1. inside out 2. upside down
3. to the moon and back 4. without hesitation
5. out of the question

4. 1. Hooray! 2. What a deal!
3. Yikes! 4. Couldn't agree more!
5. That's the tip of iceberg!

5. 1. I'm just saying'… 2. What on earth is he?
3. Not even if you paid me. 4. Just nonsense.
5. What a pity!

6. 1. put everything together 2. put your thought in the tank?
3. go Dutch? 4. share their cents?
5. put their two cents in

7. 1. God blesses her 2. Good grief
3. Good to know 4. Surprisingly
5. According to the rumors

8. 1. a bit alert 2. a bit modest
3. a bit uppity 4. irresponsible
5. absolutely moron

9. 1. there is one secret you should know 2. I know all of you don't like her
3. I am on your same page 4. since I put her name up
5. it is undebatable

10. 1. they'd better be more promising! 2. they'd be quite compromising!
3. all of you must have more choices! 4. I want all of you to work harder!
5. wish us luck on this work!

Part II: Reading Comprehension (Items 11 – 25)

Non-Fiction Part

A new report suggests Canada's Western Hudson Bay polar bear population has dropped 27 percent during the past five years.

Every autumn, bears living along the western edge of the Hudson Bay pass through the sub-Arctic town of Churchill, Manitoba. The yearly movement has helped make the area's bears one of the most studied groups in the world. Bear-watching adds about $5.3 million to the local economy each year.

But a report recently issued by the government in Canada's northern **territory** of Nunavut found that just 618 polar bears remained in the area in 2021. This represented a 50 percent drop in the population from the 1980s.

Polar bears depend on frozen salt water, called sea ice, to help them hunt seals for food. But the Arctic is now warming about four times faster than the rest of the world. Around Hudson Bay, seasonal sea ice is melting earlier in the spring, and forming later in the fall. This is forcing bears to go longer periods without food.

John Whiteman is chief research scientist at the nonprofit conservation group Polar Bears International. Speaking to Reuters news agency, he called the population drop "totally shocking." Whiteman said scientists fear that if continued sea ice loss cannot be stopped, it will one day lead to a total loss of the area's polar bear population.

(Cited from VOA)

11. What is the best headline of this article?

1. Polar Bear Population of Canada
2. Canadian Polar Bear Population Drops Sharply
3. The Reason of Polar Bear Extinction
4. The Discovery of Polar Bear Population Drop
5. Melting Sea Ice Causes Polar Bear Population Drop

12. Why frozen salt water is essential for polar bears to survive?

1. Polar bears live in water.
2. It is the source of food.
3. It helps the yearly movement of polar bears.
4. It keeps polar bears cold.
5. It helps polar beats hunt each other.

13. Which word can best replace the word "territory" in the third paragraph?

1. terrorist
2. district
3. terrifying
4. community
5. possession

Electroreception

Open your eyes in sea water and it is difficult to see much more than a murky, bleary green colour. Sounds, too, are garbled and difficult to comprehend. Without specialised equipment humans would be lost in these deep sea habitats, so how do fish make it seem so easy? Much of this is due to a biological phenomenon known as electroreception – the ability to perceive and act upon electrical stimuli as part of the overall senses. This ability is only found in aquatic or amphibious species because water is an efficient conductor of electricity.

Electroreception comes in two variants. While all animals (including humans) generate electric signals, because they are emitted by the nervous system, some animals have the ability – known as passive electroreception – to receive and decode electric signals generated by other animals in order to sense their location.

Other creatures can go further still, however. Animals with active electroreception possess bodily organs that generate special electric signals on cue. These can be used for mating signals and territorial displays as well as locating objects in the water. Active electroreceptors can differentiate between the various resistances that their electrical currents encounter. This can help them identify whether another creature is prey, predator or something that is best left alone. Active electroreception has a range of about one body length – usually just enough to give its host time to get out of the way or go in for the kill.

One fascinating use of active electroreception – known as the Jamming Avoidance Response mechanism – has been observed between members of some species known as the weakly electric fish. When two such electric fish meet in the ocean using the same frequency, each fish will then shift the frequency of its discharge so that they are transmitting on different frequencies. Doing so prevents their electroreception faculties from becoming jammed. Long before citizens' band radio users first had to yell "Get off my frequency!" at **hapless** novices cluttering the air waves, at least one species had found a way to peacefully and quickly resolve this type of dispute.

Electroreception can also play an important role in animal defences. Rays are one such example. Young ray embryos develop inside egg cases that are attached to the sea bed. The embryos keep their tails in constant motion so as to pump water and allow them to breathe through the egg's casing. If the embryo's electroreceptors detect the presence of a predatory fish in the vicinity, however, the embryo stops moving (and in so doing ceases transmitting electric currents) until the fish has moved on. Because marine life of various types is often travelling past, the embryo has evolved only to react to signals that are characteristic of the respiratory movements of potential predators such as sharks.

Many people fear swimming in the ocean because of sharks. In some respects, this concern is well grounded – humans are poorly equipped when it comes to electroreceptive defence mechanisms. Sharks, meanwhile, hunt with extraordinary **precision**. They initially lock onto their prey through a keen sense of smell (two thirds of a shark's brain is devoted entirely to its olfactory organs). As the shark reaches proximity to its prey, it tunes into electric signals that ensure a precise strike on its target; this sense is so strong that the shark even attacks blind by letting its eyes recede for protection.

Normally, when humans are attacked it is purely by accident. Since sharks cannot detect from electroreception whether or not something will satisfy their tastes, they tend to "try before they buy", taking one or two bites and then assessing the results (our sinewy muscle does not compare well with plumper, softer prey such as seals). Repeat attacks are highly likely once a human is bleeding, however; the force of the electric field is heightened by salt in the blood which creates the perfect setting for a feeding frenzy. In areas where shark attacks on humans are likely to occur, scientists are exploring ways to create artificial electroreceptors that would disorient the sharks and repel them from swimming beaches.

There is much that we do not yet know concerning how electroreception functions. Although researchers have documented how electroreception alters hunting, defence and communication systems through observation, the exact neurological processes that encode and decode this information are unclear. Scientists are also exploring the role electroreception plays in navigation. Some have proposed that salt water and magnetic fields from the Earth's core may interact to form electrical currents that sharks use for migratory purposes.

(Cited from British Council)

14. Which best expresses the main idea of the third paragraph?

1. Animals use electroreception to mate.
2. Electroreception is used to locate territory.
3. The usage of electroreception is different.
4. Animals depend on their electric signals.
5. Active electroreception is found in many marine organisms.

15. How electroreception can be used to help fish reproduce?

1. using as a mating signals
2. jamming the same frequency
3. keeping their tails in constant motion
4. identifying and locating the match
5. transmitting on different frequencies

16. Which word can best replace the word "hapless" in the fourth paragraph?

1. deliberately
2. luckless
3. opportunity
4. hopeless
5. aggressive

17. Why only creatures that live in or near water have electroreceptive abilities?

1. They need to see things in a deep-sea habitat.
2. Electricity is well conducted through water.
3. They can only perceive electrical stimuli.
4. They can emit by nervous system.
5. Electroreception is only existed in fish.

18. How electroreception might help creatures find their way over long distances?

1. Communication system through observation
2. Neurological processes
3. Territorial display and object location
4. Different frequency transmission
5. Electrical current from salt water and magnetic fields

19. Which word can best replace the word "precision" in the sixth paragraph?

1. material
2. perception
3. proactive
4. survival
5. exactness

20. What is the main purpose of the author?

1. To inform about how electroreception works
2. To illustrate how marine lives survive
3. To argue the original of electroreception
4. To inform about how fish see in the sea
5. To discuss about the creatures possessing of electroreception

Honey Bees

It was engineered to prevent fatalities from American foulbrood disease, a bacterial condition known to weaken colonies by attacking bee larvae.

The US Department of Agriculture (USDA) approved a conditional license for the vaccine this week, according to the biotech firm behind its development.

As pollinators, bees play a critical role in many aspects of the ecosystem.

The vaccine could serve as a "**breakthrough** in protecting honey bees", Dalan Animal Health CEO Annette Kleiser said in a statement.

It works by introducing an inactive version of the bacteria into the royal jelly fed to the queen, whose larvae then gain immunity.

The US has seen annual reductions in honey bee colonies since 2006, according to the USDA.

The USDA says many, sometimes overlapping, factors threaten honey bee health, including parasites, pests and disease, as well as a phenomenon called Colony Collapse Disorder, which occurs when worker bees abandon a hive and leave behind the queen.

Pollinators such as bees, birds and bats are responsible for about a third of the world's crop production, according to the United Nation's Food and Agricultural Organization.

American foulbrood disease poses a challenge for beekeepers as it is highly contagious and has no cure. The only treatment method requires burning the colony of infected bees along with the hives and equipment and treating nearby colonies with antibiotics.

The new vaccine contains an inactive version of the bacteria that causes American foulbrood disease, Paenibacillus larvae, according to Dalan Animal health.

The bacteria are incorporated into royal jelly feed given by worker bees to the queen bee, which then ingests the feed and keeps some of the vaccine in her ovaries, according to the biotech firm, which specialises in insect health and immunology.

It says this gives bee larvae immunity to the disease as they hatch and reduces death from the illness.

The new vaccine could mark an "exciting step forward for beekeepers", California State Beekeepers Association board member Trevor Tauzer said in a statement.

"If we can prevent an infection in our hives, we can avoid costly treatments and focus our energy on other important elements of keeping our bees healthy," he said.

Dalan plans to distribute the vaccine "on a limited basis" to commercial beekeepers and said the product would probably be available for purchase in the US this year.

(Cited from BBC)

21. What is the original idea in inventing the vaccine?

1. Prevent the death rate by disease of larvae
2. Protecting bee colonies
3. Increasing honey bee population
4. Weakening bee colonies
5. Introducing inactive bacteria

22. How does the vaccine work to help the bee larvae?

1. killing parasites inside honey bees
2. relieving the pain of infected bees
3. treating the bee larvae with antibiotics
4. feed directly to the bee larvae
5. feeding to the queen makes the bee larvae have immunity

23. Which word can best replace the word "breakthrough" in the fourth paragraph?

1. discover
2. break in
3. outbreak
4. scrutinize
5. protection

24. How can American foulbrood disease be treated and cured?

1. By feeding the vaccine to the honey bees
2. By burning the hives of infected bees
3. By treating with the same bacteria
4. The disease has no cure.
5. By feeding the vaccine in the queen's ovaries

25. Who is mostly affected from this scenario?

1. Farmers
2. Beekeepers
3. Honey bees
4. Bee larvae
5. The USDA

Part III: Structure and Writing (Items 26 – 40)

26. _____a breakfast for three days, I was desperate to go to the morning exercise.

1. There's not had
2. Not having
3. No haven't there
4. There's no
5. Not having had

27. My advice for negotiating the price of the service is you _____ the manager.

1. by calling
2. to call
3. calling
4. call
5. is calling

28. You dared me about this national academic exam, _____?

1. dared I not
2. daren't I
3. dared not I
4. dared you not
5. dared not you

29. Not since Jeremy found a job, _____ so happy.

1. he has been
2. has he been
3. he is
4. is he
5. he is being

30. Nolan mentioned _____ to the street market while living in the city of Bangkok.

1. having gone
2. going
3. to have gone
4. to go
5. being gone

31. I hope that, when I _____ this planet, I _____ a few people in a positive way.
1. leave / will touch
2. have left / will have touched
3. am leaving / would touch
4. leave /will have touched
5. leave / will be touching

32. _____ the intensity of the rain outside that I can't leave the school.
1. Such is
2. It's such
3. It is so
4. So is
5. Such it is

33. President Lula has been giving a televised address, _____ he called the attack in Brasilia "barbarism". and warned that everyone involved would be "found and punished".
1. in which
2. whereas
3. whom
4. which about
5. who

34. I wish he _____ to the party last night.
1. was going
2. had gone
3. went
4. had been going
5. goes

35. _____ unemployed for over four years, I _____ it difficult to get work.
1. Having been / found
2. Having / find
3. Being / find
4. Have been / have found
5. Having been / find

Directions: Choose the answer which is grammatically <u>incorrect</u>.

36. The Earth Radiation Budget Satellite, known as ERBS, was launched in 1984 **(1)** **<u>aboard</u>** the space shuttle Challenger. Although **(2)** **<u>its</u>** expected working lifetime was two years, the satellite kept **(3)** **<u>making</u>** ozone and other **(4)** **<u>atmosphere</u>** measurements until **(5)** **<u>its</u>** retirement in 2005.

37. **(1)** **<u>As</u>** concerned **(2)** **<u>were</u>** the Spanish about the **(3)** **<u>mixing</u>** of races that in 1776 **(4)** **<u>they</u>** issued legislation clamping down on **(5)** **<u>mixed-race</u>** marriages.

38. Lightning occurs in **(1)** **<u>energy</u>** storms associated with an unstable atmosphere, **(2)** **<u>requiring</u>** relatively warm and **(3)** **<u>moist</u>** air, which is **(4)** **<u>why</u>** they primarily occur in tropical latitudes and **(5)** **<u>elsewhere</u>** during the summer months.

39. The footage from December **(1)** **<u>showed</u>** a dark stain spread down **(2)** **<u>the 71-year-old</u>** president's gray trousers as he **(3)** **<u>stood for</u>** the national anthem at a road commissioning event. The video never **(4)** **<u>on air</u>** on television but **(5)** **<u>subsequently</u>** circulated on social media.

40. The journalists, who **(1)** **<u>work with</u>** the **(2)** **<u>state-run</u>** South Sudan Broadcasting Corporation, **(3)** **<u>were detained</u>** **(4)** **<u>both</u>** Tuesday and Wednesday, **(5)** **<u>said</u>** Patrick Oyet, president of the South Sudan Union of Journalists.

Part IV: Vocabulary (Items 41 – 50)

Complete the sentence with the appropriate vocabulary

When the ice caps, made of fresh water, melt, it changes many things. These changes affect the _____(41)_____ of the oceans, the currents, and the conditions for survival for myriad species. Additionally, _____(42)_____ species could move in, affecting the entire ecosystem. This has a _____(43)_____ effect, as all species are _____(44)_____ and survive according to predictable sources of food and living conditions. A specific example recently described on an environmental calendar told of the effect of global warming on polar bears. Melting ice caps _____(45)_____ bears from hunting food. Without food, they lose body fat and can be unable to give birth to cubs.

41. 1. radiation
2. ionosphere
3. salinity
4. alkalinity
5. phosphorescence

42. 1. constructive
2. practical
3. incisive
4. invasive
5. beneficial

43. 1. remarkable
2. curious
3. domino
4. singular
5. contaminants

44. 1. formulative
2. interdependent
3. dependent
4. submissive
5. altruistic

45. 1. preclude
2. encourage
3. enable
4. inspire
5. aggravate

A major _____(46)_____ problem, the magnitude of which we are just beginning to realize, is global warming. When people say that the winters are not as cold as they used to be, or that there was definitely more snow in past years, they are correct. In addition to these personal _____(47)_____, there is concrete visual evidence of global warming. Most noticeable is the _____(48)_____ of the ice caps. In recent years, glaciers have been _____(49)_____ at a greater rate than in _____(50)_____ years.

46. 1. societal
2. global
3. environmental
4. longitude
5. cyclical

47. 1. testimonials
2. denials
3. manipulations
4. denominations
5. outcries

48. 1. ramification
2. depletion
3. restitution
4. reinstatement
5. restoration

49. 1. receding
2. exacerbating
3. advancing
4. proceeding
5. progressing

50. 1. secondary
2. concluding
3. latter
4. former
5. phrase

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โครงการสอบวัดทักษะวิชาการระดับชาติ (สวช.)